APPLIED RESEARCH · CAPSTONE PRESENTATION

A Decision-Oriented Conversational Shopping Assistant

Claude does language. Deterministic code does ranking.

Preference Elicitation · Explainable Recommendations · Lifecycle Support

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Shopping Is a Decision Problem — Not a Search Problem

Why the Existing Model Breaks Down

1. ### Cognitive Overload

   Users juggle every constraint and trade-off themselves.

2. ### Vague, Evolving Needs

   People rarely know exactly what they want when they start.

3. ### Preferences Get Overwritten

   Add a new constraint — earlier choices quietly disappear.

4. ### Missing Critical Info

   Delivery time, promos, availability hidden or absent.

The Decision Burden

Falls on the user.

Decision fatigue compounds with each new option.

Users guess, settle, or abandon the purchase.

Post-purchase regret — trade-offs were never visible.

Trust erodes when recommendations have no rationale.

We built a system that addresses the decision — not just the search.

Existing Platforms Don't Support Decisions

Google Shopping	Amazon Rufus	ChatGPT Shopping	Perplexity	Our System

Capability	Google Shopping	Amazon Rufus	ChatGPT Shopping	Perplexity	Our System
Preference Elicitation	✗	Partial	Partial	✗	✓
Decision Modeling	✗	✗	✗	✗	✓
Explainable Recommendations	✗	✗	Partial	✓	✓
Delivery / Promo Info	Partial	✗	✗	✗	✓
Preference Continuity	✗	Partial	Partial	✗	✓
Lifecycle Support	✗	✗	✗	✗	✓

Our system targets every gap — in a single decision-oriented flow.

What the Assistant Actually Does

1 Understand You

- Situational context
- Smart guided Q&A
- Continuous flow

e.g. "I need something for cooking"

2 Built for You

- Personal AI guide
- Preference memory
- Multi-criteria model

remembers ≤ $80 · 3-day · quiet home

3 Clear Results

- Structured — not text heavy
- Full attributes visible
- Explainable trade-offs

shows the trade-off, not just the rank

+ Lifecycle support — value continues after the purchase.

Five Scenes · One Decision Journey

Scene 1	Scene 2	Scene 3	Scene 4	Scene 5
Frustration	**Clarification**	**Recommendations**	**Why this fits**	**Lifecycle**
User vents pain; assistant replies with empathy.	Chip questions translate vague needs into structured prefs.	12 ranked tiles — attributes visible inline.	Reasons + trade-off card + transparency banner.	Daily use, schedule, menu, post-purchase value.
Layer 1	Layer 1→2	Layer 3	Layer 4	Layer 5

Each scene corresponds to exactly one architectural layer — UI and architecture are legible to each other.

The Trust Boundary

AI handles language. Deterministic code handles the decision.

Claude (Opus 4.6) — Natural Language

Probabilistic · Translator between messy intent and structured data

① Parse preferences

② Clarifying questions

③ Explain trade-offs

④ Lifecycle suggestions

▼ TRUST BOUNDARY ▼

Rule-Based Recommendation Engine — The Decision

Deterministic · Auditable · Every weight visible · Reproducible across model updates

① Hard constraint filter

② Feasible set

③ Two-tier weighted scoring

④ Ranked output

Conceptual Layers & Runtime Components

Conceptual Flow

1. **Preference Elicitation**
2. **Decision Modeling**
3. **Recommendation & Ranking**
4. **Explainability**
5. **Lifecycle Support**

Runtime Components

React Frontend

4 stages · TypeScript · axios

Onboarding · Elicitation · Recommendations · Refinement

FastAPI Backend

/start · /answer · /recommend · /refine

Orchestrates Claude + Rule-Based Engine

Claude (Opus 4.6)

4 call sites · language only

Parse · Clarify · Explain · Lifecycle

Recommendation Engine

Hard constraints + two-tier scoring

Deterministic — no AI in ranking

Product Database

products_clean.json · 100 products · 3 categories

Smart Display · Water Bottle · Kitchen Organizer

ETL Pipeline

Playwright · Amazon + Target fallback

Normalize · Dedup · Offline batch feed

Deterministic Two-Tier Weighted Scoring

① **Hard Constraint Filter**

Budget & delivery deadline — non-negotiable.
Eliminated before scoring.

② **Build Feasible Set**

Only products satisfying all hard constraints proceed.
Typically ~100 → 10–20 candidates.

③ **Two-Tier Weighted Scoring**

General: rating · price · delivery (dynamic weights)
Feature: category-specific attributes

④ **Ranked Output**

Combined = α · General + (1 − α) · Feature, α = 0.6
Top N returned with all components preserved.

Worked Example — smart display, flexible budget, no rush

GENERAL (dynamic, from user priorities)

Rating 50%	Price 30%	Delivery 20%

FEATURE (category-specific — smart displays)

Voice 40%	Screen 30%	Smart Home 20%	Brand 10%

Status	Product	Details	Scores	Result
✓ PASS	Amazon Echo Show 8	$79 · 4.7★ · 2-day	Gen 0.82 · Feat 0.88	Combined 0.84 → #1
✓ PASS	Google Nest Hub (2nd Gen)	$65 · 4.5★ · 3-day	Gen 0.80 · Feat 0.79	Combined 0.80 → #2
✗ ELIMINATED	Amazon Echo Show 15	$249 · 4.8★ · 2-day	over $80 budget	filtered before scoring

When the system decides what to show the user, AI steps aside.

Playwright ETL · Normalization · Schema

1 Scrape
Amazon → Target

Primary + fallback

2 Normalize
Prices · ratings

Delivery → days

3 Dedup
URL exact match

+ fuzzy title prefix

4 Validate
10-field schema

TypedDict contract

5 Emit
products_clean.json

100 products · 3 cats

Normalization Examples

`"Arrives Mar 14"`	→	**`5 (days from today)`**
`"Get it in 2–3 days"`	→	**`3 (upper bound)`**
`"Ships within a week"`	→	**`7`**
`"1K reviews"`	→	**`1000`**
`"$79.99"`	→	**`79.99`**

ScrapedProduct — 10-field TypedDict

flat schema · boring · easy to reason about

- `title`
- `url`
- `image_url`
- `price`
- `rating`
- `review_count`
- `delivery_days`
- `source`
- `category`
- `fetched_at`

100 products · 3 categories · 2 active sources · two-phase dedup

Capability Validation & Scenario Walkthrough

6 / 6	4	0	100
capability gaps addressed	Claude call sites (none in ranking)	LLM inference in the ranking step	products across 3 categories

Demo — Kitchen Assistant Scenario

① **Onboarding**	'I want something for cooking' → smart display
② **Elicitation**	5 guided Qs: budget, screen, delivery, household, voice
③ **Recommendations**	Ranked list + trade-off explanation per product
④ **Refinement**	Deprioritize screen → re-rank + lifecycle ideas

Validation Findings

✓ **Information completeness**

price, rating, delivery, promos — all in one view

✓ **Interaction efficiency**

4-stage flow · zero external page navigation

✓ **Decision transparency**

every rec carries an explanation tied to priorities

✓ **Reproducibility**

identical inputs → identical ranking

Heuristic self-assessment against the six capability gaps · formal user study is future work

Three End-to-End Scenarios — One Per Category

Kitchen Assistant

Sarah · 38 · busy parent

Opening prompt

"I cook every day but my phone screen is too small to follow recipes."

Chip picks

Cooking · Under $150 · No rush

Top recommendation

Echo Show 8 / Echo Show 5 with Smart Bulb

Lifecycle value

Hands-free recipes · meal-plan dashboard · family hub

Gym Hydration

Mike · 27 · runner

Opening prompt

"My bottle is heavy, leaks, and I can't track my water."

Chip picks

Gym · Insulated · Lightweight

Top recommendation

Owala FreeSip 19oz Insulated ($14.99)

Lifecycle value

Hydration tracker · workout reminders · flavor ideas

Cabinet Reset

Priya · 31 · small apartment

Opening prompt

"My drawers are chaos. Every morning I waste time hunting."

Chip picks

Cabinets · Not enough space · Stackable

Top recommendation

Brightroom Stackable Pantry Bin Set ($9–16)

Lifecycle value

Weekly tidy nudges · smart restock · calm counters

Same five-layer architecture · same UI flow · three product categories — generalisability demonstrated.

What This Proves · What It Doesn't · What's Next

Limitations

- **Small dataset**
  100 products · 3 categories only
- **No user study**
  heuristic evaluation only
- **Static data**
  prices & delivery change dynamically
- **Rule-based weights**
  inferred, not learned

Conclusions & Impact

- **Decision support feasible**
  end-to-end with Claude Opus 4.6
- **Structured elicitation**
  designed to address cognitive load
- **Explainable trade-offs**
  transparent, reproducible rankings
- **ETL validates the approach**
  Playwright scrape · normalize · dedup
- **Lifecycle value**
  extends past the purchase itself

Future Work

- **Formal user study**
  SUS + task completion time
- **Live retailer API**
  real-time pricing
- **Learned ranking**
  from interaction data
- **More categories**
  beyond the current three
- **Mobile-first UI**
  redesign for small screens

Thank You

Questions & Discussion

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Project	A Decision-Oriented Conversational Shopping Assistant
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